

Linux Commands

Beginner's Guide for Kali Linux

Easy Explanations • Real Analogies • Hands-On Exercises

Section 1: File Operation Commands

These are the commands you will use every day to work with files and folders in Kali Linux. Each command below includes a simple explanation, a relatable analogy, the most useful flags, and real examples you can type directly into your terminal.

cat

What it does: Reads and displays a file's content on the screen. Think of it as opening a file and reading it out loud to the terminal.

Beginner Analogy: Like opening a book and reading every page aloud — you can see everything, but you are not changing anything.

Flag	What it does
-n	Show line numbers next to every line (useful for debugging code)
-b	Show line numbers only for lines that have text (skips blank lines)

```
# Display a file
cat hello.txt

# Display with line numbers
cat -n hello.txt

# Combine two files into one new file
cat file1.txt file2.txt > combined.txt
```

cp

What it does: Copies a file or folder from one place to another. The original file stays where it is — cp just makes a duplicate.

Beginner Analogy: Like using a photocopier — the original stays on your desk, and a new copy goes somewhere else.

Flag	What it does
-r	Copy entire folders (including everything inside them)
-i	Ask you before overwriting a file that already exists (safe mode!)

```
# Copy one file to another name
cp original.txt copy.txt

# Copy a whole folder
cp -r myfolder/ backup_folder/

# Copy safely (ask before overwriting)
cp -i report.txt backup_report.txt
```



What it does: Moves a file to a new location OR renames it. Unlike cp, the original file is removed from its old location.

Beginner Analogy: Like picking up a book from one shelf and placing it on another shelf — it is no longer where it was before.

Flag	What it does
-i	Ask before overwriting any existing file at the destination
-n	Never overwrite files — if a file already exists there, skip it

```
# Rename a file
mv oldname.txt newname.txt

# Move a file to a different folder
mv file.txt /home/user/Documents/

# Move safely
mv -i draft.txt final.txt
```



rm

What it does: Permanently deletes files or folders. **IMPORTANT:** Linux has no Recycle Bin — once deleted, files are gone forever!

Beginner Analogy: Like shredding a paper document — once through the shredder, you cannot get it back.

Flag	What it does
-r	Delete a folder and everything inside it (recursive)
-f	Force delete — no questions asked, no error messages

```
# Delete one file
rm unwanted.txt

# Delete an entire folder
rm -r old_project/

# WARNING: This deletes EVERYTHING inside, with no confirmation
rm -rf dangerous_folder/
```

Warning: NEVER run `rm -rf /` — this deletes your entire system! Always double-check the path before using `rm -rf`.

touch

What it does: Creates a new empty file instantly. If the file already exists, it just updates the file's 'last modified' time.

Beginner Analogy: Like placing a blank sticky note on a board — the note now exists, but nothing is written on it yet.

Flag	What it does
-c	Do not create the file if it does not already exist
-t	Set a specific date and time for the file (format: YYYYMMDDhhmm)

```
# Create a new empty file
touch newfile.txt

# Create multiple files at once
touch file1.txt file2.txt file3.txt

# Set a custom timestamp
touch -t 202605261200 log.txt
```

diff

What it does: Compares two files and shows you the differences between them, line by line.

Beginner Analogy: Like placing two essays side by side and highlighting the sentences that are different.

Flag	What it does
-u	Unified format — shows a few lines of context around each change (easier to read)

```
# Compare two files
diff file1.txt file2.txt

# Compare with context lines (easier to read)
diff -u old_config.conf new_config.conf
```

tar

What it does: Bundles multiple files into one archive file (like a zip). Can also compress the archive to make it smaller.

Beginner Analogy: Like packing several books into one box (tar) and then shrink-wrapping the box to save space (gzip).

Flag	What it does
-c	Create a new archive
-x	Extract (unpack) an existing archive
-v	Verbose — show which files are being processed
-f	Specify the archive filename
-z	Also compress using gzip (makes the file smaller)

```
# Pack files into one archive
tar -cvf myarchive.tar file1.txt file2.txt

# Pack AND compress (smaller file size)
tar -czvf backup.tar.gz /home/user/Documents/

# Unpack (extract) an archive
tar -xvf myarchive.tar
```

gzip / gunzip

What it does: Compares and compresses a file to make it smaller (gzip), or decompresses it back to original (gunzip).

Beginner Analogy: Like squeezing air out of a storage bag — the contents are the same, just taking less space.

Flag	What it does
-d	Decompress a file (same as using gunzip)
-k	Keep the original file after compressing (do not delete it)

```
# Compress a file (creates filename.txt.gz)
gzip filename.txt

# Decompress it back
gunzip filename.txt.gz

# Compress but keep the original file too
gzip -k report.txt
```

unzip

What it does: Extracts files from a .zip archive. Very common for files downloaded from the internet.

Beginner Analogy: Like opening a wrapped gift box and taking everything out.

Flag	What it does
-l	List what is inside the zip file WITHOUT extracting it

```
# Extract a zip file
unzip archive.zip

# See what is inside without extracting
unzip -l project.zip
```

Section 2: System Management Commands

These commands help you manage your Kali Linux system — navigate folders, control users, monitor performance, and more. Use this as a quick reference cheat sheet.

Folder Navigation

Command	What it does (simple)
ls	List all files in current folder
cd foldername	Go into a folder
pwd	Show your current location
mkdir newfolder	Create a new folder
rmdir emptyfolder	Remove an empty folder

Permissions

Command	What it does (simple)
chmod 755 file	Change who can read/write/run a file
chown user file	Change who owns a file
chgrp group file	Change the group that owns a file

Users & Accounts

Command	What it does (simple)
useradd john	Create a new user named john
passwd john	Set a password for john
whoami	See who you are currently logged in as

Running Processes

Command	What it does (simple)
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ps aux	List all running programs
top	Live view of CPU/memory usage
kill 1234	Stop process with ID 1234
htop	Interactive process manager (nicer than top)

Networking

Command	What it does (simple)
ping google.com	Test if you can reach a website
ifconfig	Show your IP address and network info
ssh user@ip	Remotely log into another machine
curl http://site.com	Fetch content from a URL
netstat -tulnp	See all open network ports

Disk & Memory

Command	What it does (simple)
df -h	Show how much disk space is used
du -sh folder/	Show how big a folder is
free -h	Show RAM usage
uptime	How long the system has been running

Search & Text Tools

Command	What it does (simple)
grep 'word' file	Search for a word inside a file
find / -name file.txt	Find a file anywhere on the system
awk '{print \$1}' file	Extract columns from text
sed 's/old/new/g' file	Replace text inside a file

Scheduling & Packages

Command	What it does (simple)
<code>crontab -e</code>	Edit scheduled automatic tasks
<code>at 23:30</code>	Run something once at a specific time
<code>sudo apt install pkg</code>	Install a package (Kali/Debian)
<code>sudo apt update</code>	Refresh list of available packages

Section 3: Practice Exercises with Solutions

Try each exercise on your own first, then read the solution and explanation. All exercises can be run directly in Kali Linux terminal.

Exercise 1: Count Visitor IPs from a Web Log

Task: Read access.log, find unique IP addresses, count how many times each appeared, and sort from most to least.

```
awk '{print $1}' access.log | sort | uniq -c | sort -nr
```

What this means: awk grabs the first word (the IP) from each line. sort groups matching IPs together. uniq -c counts how many times each IP appeared. sort -nr puts the highest count at the top.

Exercise 2: Replace http with https in a Config File

Task: Automatically change every 'http' link to 'https' inside config.txt.

```
sed -i 's/http/https/g' config.txt
```

What this means: sed is a stream editor. -i means edit the file directly. s/http/https/g means: substitute every (g = global) occurrence of 'http' with 'https'.

Exercise 3: Find ERRORS in Log Files

Task: Search all files in /var/log and show every line that contains the word ERROR, with color highlighting.

```
grep -r --color=auto "ERROR" /var/log
```

What this means: grep searches for matching text. -r means search recursively (all subfolders too). --color=auto highlights the matching word so it is easy to spot.

Exercise 4: Find Files Larger Than 100MB

Task: Scan /home/user and list every file that is bigger than 100 megabytes.

```
find /home/user -type f -size +100M
```

What this means: find searches your file system. -type f means files only (not folders). -size +100M means bigger than 100 megabytes.

Exercise 5: Schedule an Automatic Daily Backup

Task: Run /home/user/backup.sh every single day at exactly 2:00 AM automatically.

```
0 2 * * * /home/user/backup.sh >> /home/user/backup.log 2>&1
```

What this means: This is a cron job entry. 0 2 * * * means: minute 0, hour 2, every day. >> appends output to a log file. Add it by running: crontab -e.

Exercise 6: Schedule a One-Time Shutdown at 11:30 PM

Task: Queue a system shutdown tonight at 23:30 using the 'at' scheduler.

```
echo "shutdown -h now" | at 23:30
```

What this means: at schedules a one-time command. echo sends the shutdown command into at, which runs it at 23:30 tonight. Unlike cron, it does not repeat.

Exercise 7: Install Packages on Kali Linux

Task: Update your package list and install the curl tool (Kali/Debian) or htop (RedHat/CentOS).

```
sudo apt update && sudo apt install curl -y
```

What this means: apt is the package manager for Kali Linux (Debian-based). update refreshes the list of available software. install curl actually installs it. -y means 'yes to all prompts'.

Exercise 8: Find Which Process is Using Port 22

Task: Check which program is currently listening on SSH port 22.

```
sudo lsof -i :22
```

What this means: lsof stands for 'list open files'. -i :22 filters to show only connections on port 22. You will see the program name and its process ID (PID).

Exercise 9: Watch System Calls Made by ls

Task: Trace every low-level system action triggered when you run the ls command.

```
strace ls 2>&1 | grep open
```

What this means: strace records every system call a program makes. 2>&1 merges error output with normal output. grep open filters to show only file-open actions — useful for debugging.

Exercise 10: Clone a Hard Drive Bit by Bit

Task: Copy everything from /dev/sda to /dev/sdb in safe 4MB chunks, showing live progress.

```
sudo dd if=/dev/sda of=/dev/sdb bs=4M status=progress
```

What this means: dd is a low-level copy tool. if = input (source disk). of = output (destination disk). bs=4M means copy 4 megabytes at a time (faster). status=progress shows live speed and progress.

Exercise 11: Mount a USB Drive, Check Space, Then Safely Eject

Task: Attach `/dev/sdb1` to `/mnt/usb`, check available space, then safely remove the USB drive.

```
sudo mount /dev/sdb1 /mnt/usb && df -h | grep /mnt/usb
sudo umount /mnt/usb
```

What this means: `mount` makes the USB accessible at `/mnt/usb`. `df -h` shows disk usage in readable units (GB/MB). Always run `umount` before physically unplugging — skipping this can corrupt your files!



QUICK TIPS & GOLDEN RULES FOR BEGINNERS

- Always read the command twice before pressing Enter — especially with **rm**!
- Use the **Tab key** to auto-complete file and folder names.
- Press **Ctrl+C** to cancel a running command at any time.
- Use the **Up arrow key** to recall your previous commands.
- Type `'man commandname'` (e.g., `man ls`) to read the full manual for any command.
- When in doubt, add the `-i` flag to `cp` and `mv` to avoid accidental overwrites.
- Use `'sudo'` before commands that need admin/root privileges.
- Always `umount` a USB drive before physically removing it.